

Can Concept Erasure Reproduce Concept Exclusion? A Matched Evaluation of EMBER, RMU, and SNMF

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Study design schematic to draw from final protocol

Figure 1: Planned comparison of the full model, concept-excluded twin, and three erasure methods on identical target, neighboring, and unrelated sentence completions.

Abstract

Plan: Target 150–190 words. State the erasure question; matched full/twin/ EMBER/RMU/SNMF design; three concepts and three evaluation groups; answer NLL and full-vocabulary KL; the most important result and qualification once known. Write last. No numbers until all five models use the same held-out protocol.

1 Introduction

Plan: About 0.7 page. Motivate selective concept erasure and explain why “forgetting” needs two checks: approaching the concept-excluded twin on the target and retaining correct-answer support elsewhere. Define the full model, excluded twin, and post-hoc edit before using those terms repeatedly. Name Ancient Rome, Baseball, and AI. State that the study measures answer-token probability and next-token distributional similarity, not generated-answer accuracy. End with 2–3 concrete contributions once results exist.

2 Background and Related Work

2.1 Concept Exclusion as an Empirical Reference

Plan: About 0.2 page. Explain loss masking during training, the operational blacklist, and why the resulting twin is a reference for this intervention, not proof that all knowledge of the concept is absent. Add citations.

2.2 Post-hoc Erasure Methods

Plan: About 0.3 page. Introduce EMBER’s embedding edit, RMU’s MLP update, and SNMF’s MLP feature edit in comparable terms. Explain what parameters each can change and cite the original methods. Save implementation specifics for Section 3.

Plan: About 0.3 page. State architecture, corpus, checkpoint, initialization, batch order, optimizer, duration, and how labels from selected chunks are masked. Explain the shared full model and one twin per concept.

3 Experimental Design

3.1 Shared Base Model and Matched Twins

Plan: About 0.3 page. State architecture, corpus, checkpoint, initialization, batch order, optimizer, duration, and how labels from selected chunks are masked. Explain the shared full model and one twin per concept.

3.2 Target Concepts and Exclusion Scope

Plan: About 0.25 page. Define Ancient Rome, Baseball, and AI blacklist rules; give QID/chunk footprint and any precision/coverage audit. Explicitly connect the target evaluation group to each exclusion scope.

3.3 Erasure Checkpoints and Model Selection

Plan: About 0.35 page. State that EMBER, RMU, and SNMF start from the same full checkpoint. Summarize edited parameter families, training or factorization data, and how settings were selected without using final held-out items. Describe one selected checkpoint per method and concept. Put exact settings and chronology in Appendix A.

3.4 Evaluation Data Separation

Plan: About 0.2 page. Explain separate erasure-training, selection, and final evaluation sentence completions, identical across all five models. Introduce target, neighboring, and unrelated groups. Every item has one fixed correct continuation. Counts and question provenance go in Appendix B.

4 Evaluation Protocol

4.1 Three Groups and Fixed Correct Answers

Plan: About 0.2 page. Define target, neighboring, and unrelated groups and their distinct roles. Each item has a declarative sentence-completion prompt and one fixed correct continuation, for example “The city of Rome is located in” followed by “Italy.” Score only the answer tokens. State explicitly that no candidates are ranked and no text is generated.

Concept	Chunks	Target Q	Neighbor Q	Unrelated Q
Ancient Rome	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Baseball	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AI	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 1: Planned study inventory. Replace placeholders with final masked chunk and held-out completion counts.

4.2 Answer NLL Difference

Plan: About 0.25 page. For each question, average the negative log-probability of each correct-answer token given the prompt and preceding correct-answer tokens. Subtract the reference model’s per-question NLL from the evaluated model’s, then average question differences. Positive means less probability on the correct answer. A signed mean near zero may conceal cancellation.

4.3 Teacher-Forced Full-Vocabulary KL

Plan: About 0.25 page. At each correct-answer position, compare the full next-token vocabulary distributions under identical teacher-forced prefixes. Compute $KL(\text{reference} \rightarrow \text{evaluated})$, average answer positions within each question, then average questions. KL measures distributional similarity, not whether the correct answer improved, and is not divergence between generated answers.

4.4 Model Comparisons and Uncertainty

Plan: About 0.3 page. Twin minus full establishes the exclusion effect; erasure minus twin measures resemblance; erasure minus full reveals target change and preservation. Report both metrics separately for target, neighboring, and unrelated groups. Define identical prompts, answers, tokenization and context settings. State paired bootstrap unit, paraphrase clustering and intervals; avoid equivalence claims from a near-zero mean.

5 Results

5.1 What Concept Exclusion Changes

Plan: About 0.4 page. For each concept, establish twin-minus-full answer NLL and $KL(\text{full} \rightarrow \text{twin})$ on target questions, then neighboring and unrelated questions. State where the exclusion reference has weak observable effect. Point to complete per-group twin rows in Appendix D.

5.2 How Closely Erasure Matches the Twin on Target Items

Plan: About 0.7 page. Compare EMBER, RMU and SNMF with the twin and full model on target answer NLL and full-vocabulary KL. Show paired intervals and inspect per-question variation; a near-zero signed mean NLL does not establish identical behavior. Discuss substantive differences

Signed answer-NLL changes by concept, model, and question group

Figure 2: Planned answer-NLL differences from the full model on target, neighboring, and unrelated sentence completions, with paired intervals.

among Rome, Baseball, and AI only after the new measurements are filled.

5.3 Neighboring and Unrelated Correct-Answer Support

Plan: About 0.6 page. Give neighboring and unrelated answer-NLL and KL comparisons separately for every method. Positive NLL from full indicates weaker prediction of the correct answer; KL indicates a distributional change but has no direction of quality. Discuss neighboring spillover and unrelated preservation relative to both full and twin.

5.4 Where Answer NLL and KL Disagree

Plan: About 0.3 page. Examine cases where the correct-answer probability appears stable but the full next-token distribution changes, or vice versa. Check whether a near-zero mean signed NLL difference hides opposing question-level changes. Use concrete paired examples only if supported by the final data.

6 Analysis and Discussion

6.1 What Counts as Matching Concept Exclusion?

Plan: About 0.35 page. Synthesize target similarity and preservation across answer NLL and KL. Explain why higher target answer NLL than the twin does not establish a better edit, and why low KL alone cannot establish knowledge preservation. Respect the scope of fixed-answer sentence completions.

6.2 Why Concepts and Metrics May Disagree

Plan: About 0.45 page. Investigate supported explanations: blacklist coverage, neighboring-topic overlap, per-question NLL cancellation, and disagreements between correct-answer NLL and full-vocabulary KL. Clearly distinguish observed evidence from hypotheses. Add a small diagnostic figure only

Concept	Erasure model	Δ NLL vs full	Δ NLL vs twin	KL(twin $\rightarrow$ erasure)
Rome	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Baseball	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AI	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 2: Target-question summary to populate after common-protocol scoring. NLL changes are signed evaluated-minus-reference differences. Show paired intervals in the final version and the twin-minus-full reference effect nearby.

if it clarifies a central result; otherwise put diagnostics in Appendix E.

7 Limitations

Plan: About 0.4 page. State limits from operational blacklist coverage, fixed correct answers rather than generated-answer accuracy, no general passage-prediction measurement, single training seed/checkpoint per arm, finite and potentially dependent questions, and teacher-forced KL. Identify any exploratory selection or unequal evaluation cell honestly when results are filled.

8 Conclusion

Plan: About 0.2 page. Answer which method most closely matches each twin on correct-answer NLL and full-vocabulary KL, and whether neighboring/unrelated answer support is preserved, at the level actually supported. Do not claim generated-answer accuracy or passage-level erasure. One restrained next step.

AI Disclosure and Reflection

Plan: About 0.2 page. Fill with the actual AI tools and models the team used, their roles in coding, editing, or analysis, and a concise reflection on verification. The course guidelines require specific names; do not guess.

A Corpus Exclusion and Model Configurations

Plan: Table A1: one row per concept and method with checkpoint ID, erasure settings, training/selection data, edited parameters, and seed. Give exact blacklist rules and audit estimates. Do not paste raw run logs.

B Sentence-Completion Evaluation Sets

Plan: Table A2: target, neighboring, and unrelated question counts for each concept; item construction and exclusions; one representative fixed prompt/answer; question provenance and train/selection/test overlap checks.

C Scoring and Statistical Details

Plan: Write exact equations for per-question answer-token NLL, signed evaluated-minus-reference group mean, and full-vocabulary KL(reference $\rightarrow$ evaluated) under correct-answer teacher forcing. Specify tokenization, prefix, answer-token mask, position/question aggregation, bootstrap resampling and paraphrase clustering. Include one worked item if useful. No option-accuracy or passage-NLL metrics belong in this protocol.

D Complete Results by Concept

D.1 Ancient Rome

Plan: Table A3: all five models, all three question groups, answer NLL and full-vocabulary KL; underlying group means, full/twin contrasts, and paired intervals. Split by metric if needed.

D.2 Baseball

Plan: Table A4 with the identical structure and question-level protocol.

D.3 Artificial Intelligence

Plan: Table A5 with the identical structure and question-level protocol.

E Diagnostic Analyses

Plan: Optional: per-question signed answer-NLL differences showing possible cancellation; examples where NLL and KL disagree; distributional plots and supported case studies. Include only analyses actually performed and needed to interpret main results.

F Reproducibility Notes

Plan: Final scoring script/version, model IDs, seeds, exact question manifest and split chronology, tokenization/context settings, answer scoring masks, and any deviations from the common protocol. Keep this concise.